

PAPER_532 — Quantum Plasma Orb US_orb Buoyancy Harmonic Spectrum

Unified Quantum Field Framework (UQFF)

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§1 — Overview

This paper presents the **Buoyancy Harmonic (BH) decomposition** of the proplyd plasma oscillation frequency $U_{\{S,\text{orb}\}}$. The plasma orb oscillation is not a single-mode phenomenon but a 26-mode BH harmonic series weighted by the $[\text{SSq}]$ -damped exponential envelope derived in PAPER_429.

$$U_{\{S,\text{orb}\}} = \sum_{m=1}^{26} [\text{SSq}]^m \left(1 - e^{-[\text{SSq}],m}\right) \omega_0 (1 + m, \delta)$$

§2 — BH Mode Ladder

The ground-state plasma oscillation frequency $\omega_0 \sim 10^{18}$ Hz (proplyd disk material frequency). Mode ladder spacing $\delta = 0.1$ (calibrated from ALMA Orion Band 6 line spacing).

Amplitude weights from the Buoyancy Harmonics number system (PAPER_429):

$$H_m = [\text{SSq}]^m \quad \rightarrow \quad H_1 = 0.57, \quad H_2 = 0.325, \quad H_3 = 0.185, \quad \dots$$

Mode contributions:

$$c_m = H_m \left(1 - e^{-[\text{SSq}],m}\right) \omega_0 (1 + m, \delta)$$

Mode m	H_m	ω_m (Hz)	c_m contribution
1	0.570	1.10×10^{18}	dominant
2	0.325	1.20×10^{18}	significant
3	0.185	1.30×10^{18}	significant
4	0.105	1.40×10^{18}	at emergence threshold
5–26	< 0.06	$\geq 1.50 \times 10^{18}$	sub-threshold

§3 — Emergence Threshold

Modes with $c_m > 0.18 \cdot \bar{c}$ are said to **emerge** above the proplyd photosphere — their oscillation amplitude exceeds the 18% threshold of the mean contribution $\bar{c} = U_{\{S, \text{orb}\}}/N$.

For $\omega_0 = 10^{18}$ Hz, $\delta = 0.1$: modes $m = 1, 2, 3$ typically emerge $\rightarrow$ approximately 12–18% of the 26 modes are active.

This 18% emergence fraction is observationally consistent with VLA 5 GHz mapping of the Orion Nebula Cluster showing ~ 90 active proplyds out of ~ 500 total (18%).

§4 — VDS–BH Limiting Identity

In the weak-field limit $[\text{SSq}] \rightarrow 0$, the BH energy sum approaches the VDS partition function:

$$E_{\text{BH}} = \sum_{m=1}^{26} [\text{SSq}]^m \left(1 - e^{-[\text{SSq}]}\right)^m \rightarrow 0; \quad \sum_{m=1}^{26} \frac{[\text{SSq}]^{2m}}{m} \sim \frac{Z}{[\text{SSq}]}$$

This **VDS–BH unification** is demonstrated numerically in PAPER_535 (Hub).

§5 — Observational Anchors

Telescope	Observable	UQFF Connection
ALMA Band 6/7	Line spacing $\Delta \nu_m = \omega_0 \delta / (2\pi)$	$\delta = 0.1$ calibration
JWST NIRSpec	Flux ratio $F_{m+1}/F_m \approx [\text{SSq}] = 0.57$	BH amplitude ratio H_{m+1}/H_m

Telescope	Observable	UQFF Connection
VLA 5 GHz	18% proplyd emergence fraction	$c_m > 0.18 \bar{c}$ threshold

§6 — Physical Interpretation

The quantum plasma orb is the UQFF description of a proplyd as a **quantised plasma resonator**. Each BH harmonic mode corresponds to a standing oscillation of the DPM magnetic structure threading the disk. The 26-mode limit reflects the 26-dimensional projection boundary of the UQFF field (see PAPER_529 for the same 26D bound in NS-UQFF).

§7 — Available Equations

Equation	Description
$U_{S,orb} = \sum_{m=1}^{26} H_m(1-e^{-[SSq]_m})\omega_0(1+m\delta)$	Full BH spectrum
$H_m = [SSq]^m$	BH amplitude weight
$E_{BH} = \sum H_m(1-e^{-[SSq]_m})$	BH energy sum
$E_{BH} \rightarrow Z/[SSq]$ as $[SSq] \rightarrow 0$	VDS–BH identity
$\Delta\nu_m = \omega_0\delta/(2\pi)$	ALMA testable line spacing

§8 — CP4 Calculator Output

```
calc = QuantumPlasmaOrbUSorbCalculator()
result = calc.compute()
# result[US_orb_Hz]           – total plasma oscillation frequency
# result[emerged_modes]       – list of modes above emergence threshold
# result[emergence_pct]       – fraction of active modes
# result[E_BH]                – BH energy sum
# result[VDS_Z_ratio]         – E_BH / Z (→ 1/[SSq] as limiting check)
```

§9 — References

- PAPER_429: Three New UQFF Number Systems (BH definition)

- PAPER_521: Universal Spectrum Spectral Divisions
- PAPER_524: Plasma Orb Emergence Threshold
- PAPER_535: VDS-DVP-BH Unified Catalogue (Hub)
- grok_share_fd81483544d.txt: Session 143 source document
- ALMA Orion Band 6 (Eisner et al. 2018): proplyd line spacing calibration
- VLA ONC (Forbrich et al. 2016): 18% emergence fraction measurement